Sahaj Bhadja

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Summary

Software analyst with 2+ years delivering **full-stack web applications**, **test automation**, and **data-driven troubleshooting** across SQL-backed systems, CI/CD pipelines, and real-time operational workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Built **data-intensive services and user interfaces** for 500+ users, reducing latency from **1.2s to 350ms** via SQL optimization, API tuning, and instrumentation-driven diagnostics.
- Validated **5K+ monthly workflow executions** with fault checks and rollback safeguards, improving defect prevention, deployment confidence, and post-release support reliability.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Developed **React + Flask application features** handling 10K+ daily requests, translating requirements into maintainable UI/API solutions with clear acceptance and test criteria.
- Improved **CI/CD and regression coverage** using automated quality gates and structured defect triage, reducing release turnaround by **60%** while maintaining product stability.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Built **pipeline-based validation workflows** over 100K+ trajectory logs, enabling reproducible functional and performance testing across complex real-time behavior scenarios.
- Created **technical bug diagnostics and reports** that accelerated issue communication between researchers and engineers and shortened root-cause analysis cycles.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Engineered **API/ETL processing services** with Python/Java, AWS Glue, and S3, improving response times by **30%** and strengthening cross-system reliability.
- Maintained **20+ automated test suites** using Jenkins, PyTest, and Mockito, increasing pipeline efficiency by **68%** and reducing recurring production defects.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Implemented **MERN + WebSocket modules** supporting eight workflows and 2M+ synchronized records, improving consistency, monitoring visibility, and responsiveness for users.
- Automated **Docker/Jenkins deployment and smoke tests**, reducing rollout time by **50%** and improving release quality across staging and production environments.

Skills

Languages: **C#**, **Python**, C++, JavaScript/TypeScript, SQL, Go basics
Web Stack: HTML5, CSS3, Angular/React, REST APIs, MVC patterns, responsive UI development
Testing & QA: functional/regression/integration testing, defect tracking, troubleshooting, usability checks
Data & Cloud: Microsoft SQL Server, relational design, Azure/AWS/GCP familiarity, deployment pipelines
Tools: Visual Studio, VS Code, Git/TFVC, Jenkins, Docker, CI/CD, Agile team collaboration

Projects

Smart Traffic Operations Dashboard | *Angular, .NET API, SQL Server, Azure*
- Built a **real-time operations dashboard** for signal and incident views with alerting and role-based APIs, improving troubleshooting speed and operator visibility during peak traffic events.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*
- Developed a **scalable full-stack platform** supporting 50K+ users and 1M+ transactions, with secure service integration and resilient deployment practices in cloud environments.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Designed a **schema-aware SQL assistant** across 10+ datasets with validation checks and explainable outputs, improving analyst productivity and reducing query debugging effort.